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QUESTIONS

1. In an approximation, the difference between the exact solution (x_{true}) and the approximate solution ($x_{approximate}$) is called the *true absolute error* and its formula is: $E_t = |x_{true} - x_{approximate}|$. However, you have learnt a second type of error: *relative error* ($E_r = \left|\frac{x_{true} - x_{approximate}}{x_{true}}\right|$). Why might we need to use relative error? Please explain with a simple example that we use relative error in numerical calculations to show that the approximation is more meaningful/better. (10 pts)

2. Let the function $f(x)$ be continuous on the interval $[a, b]$. How can you guarantee that the equation $f(x) = 0$ has at least one real root in the interval $[a, b]$? Explain briefly. (10 pts)

3. Let $f(x) = \frac{e^x}{1+x^2}$ and $x_0 = 0$. Determine the Taylor expansion about x_0 with the first three terms. Calculate the approximate value of the $f(x)$ function by taking $x = 0.125$ in the expansion you obtained. (15 pts)

4. Consider the equation $x - cos(x) = 0$.
   a. Prove that this equation has at least one real root in the interval $[1, 2]$. (5p).
   b. Calculate the approximate root of the given equation within the interval $[1, 2]$ using the Regula Falsi method after two iterations. (10p)

5. Find the root of the equation $\sqrt{x} + x^2 = 7$ using Newton's method. Start at $x = 7$ and carry out the first five iterations. (20 pts)

6. Consider this function: $f(x) = x^3 - x - e^x - 2$

   An approximate solution will be calculated using the fixed-point iteration method with the initial condition $x_0 = 2$.
   a. For this function; find the iteration formula using the initial condition. (Note: This function has four different iteration formula candidates. Please write and explain all.) (20p)
   b. Using this iteration formula, calculate the approximate root after 2 iterations. (10p)